Gas-Phase and Heterogeneous Equilibria

Answer key with worked solutions

A. Writing K

1. Kp expressions.

> (a) $K_\mathrm{p} = \dfrac{(P_{\mathrm{SO_3}})^2}{(P_{\mathrm{SO_2}})^2\,P_{\mathrm{O_2}}}$ (b) $K_\mathrm{p} = \dfrac{(P_{\mathrm{NO_2}})^2}{(P_{\mathrm{NO}})^2\,P_{\mathrm{O_2}}}$
> (c) $K_\mathrm{p} = P_{\mathrm{H_2O}}$ (the vapour pressure; liquid left out) (d) $K_\mathrm{p} = P_{\mathrm{NH_3}}\,P_{\mathrm{HCl}}$ (solid left out)

2. Δn.

> (a) **−1**: Kp smaller than Kc (b) **−1**: smaller (c) **+1**: Kp larger than Kc (d) **+2**: larger. At 25 °C, RT = 24.47, which is greater than 1, so a positive Δn makes Kp larger and a negative Δn makes it smaller.

B. Calculating Kp and Kc

3. Kp from pressures.

> $$K_\mathrm{p} = \frac{(1.2)^2}{(0.20)^2 \times 0.10} = \frac{1.44}{0.0040} = \mathbf{360}$$

4. Kc to Kp.

> $RT = 0.08206\ \mathrm{L\ atm/(mol\ K)} \times 1000\ \mathrm{K} = 82.06\ \mathrm{L\ atm/mol}$; Δn = −1
> $K_\mathrm{p} = 280 \times (82.06)^{-1} = \mathbf{3.41}$

5. Kp to Kc.

> $RT = 0.08206\ \mathrm{L\ atm/(mol\ K)} \times 298.15\ \mathrm{K} = 24.47\ \mathrm{L\ atm/mol}$; Δn = +1
> $$K_\mathrm{c} = \frac{K_\mathrm{p}}{(RT)^{\Delta n}} = \frac{0.15}{24.47} = \mathbf{6.1 \times 10^{-3}}$$

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C. Q and ICE tables

6. Q and direction.

> $Q = \dfrac{(0.50)^2}{0.10 \times 0.10} = \mathbf{25}$. Q (25) is less than K (50), so the reaction goes **forward**, making more HI.

7. ICE table.

> ICE (atm): N_2O_4: $1.00 - x$; NO_2: $2x$.
>
> $$\frac{(2x)^2}{1.00 - x} = 0.15$$
> $$4x^2 + 0.15x - 0.15 = 0$$
> $$x = \frac{-0.15 + \sqrt{0.15^2 + 4 \times 4 \times 0.15}}{2 \times 4} = 0.176$$
>
> $P_{NO_2} = 2 \times 0.176 = \mathbf{0.352\ atm}$; $P_{N_2O_4} = 1.00 - 0.176 = \mathbf{0.824\ atm}$. Check: $(0.352)^2 \div 0.824 = 0.150$ ✓

D. Challenge

8. Limestone in a sealed container.

> Kp = $P(CO_2)$, which is fixed at a given temperature. (a) Adding more solid $CaCO_3$ does **not change** $P(CO_2)$: solids are not in the K expression. (b) Removing CO_2 makes Q less than K, so more $CaCO_3$ decomposes until **$P(CO_2)$ returns** to the value of Kp.

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